

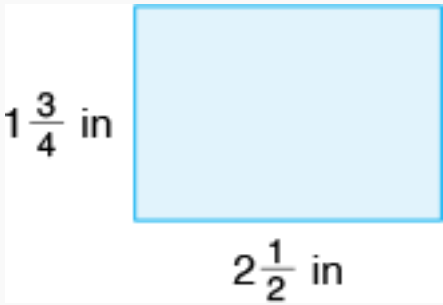
Problems with Mixed Numbers

When you are multiplying or dividing, you should always change mixed numbers to improper fractions.

Example 1

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A rectangle measures $1\frac{3}{4}$ inches by $2\frac{1}{2}$ inches. What is the area of the rectangle in square inches? (*Hint:* To find the area of a rectangle, multiply the length by the width.)



Step 1

Change mixed numbers to improper fractions.

$$1\frac{3}{4} = \frac{7}{4} \qquad 2\frac{1}{2} = \frac{5}{2}$$

Step 2

Multiply the improper fractions.

$$\frac{7}{4} \times \frac{5}{2} = \frac{35}{8}$$

Step 3

Change the improper fraction to a mixed number.

$$\frac{35}{8} = 4\frac{3}{8}$$

The area of the rectangle is $4\frac{3}{8}$ square inches.

As you work, remember that your final answer should be in lowest terms. Always check the fraction part of the answer to see if it can be simplified.

Example 2

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Bill used $2\frac{1}{2}$ gallons of paint to cover one side of a warehouse. Each side is the same size. How many sides could he paint if he had $6\frac{1}{4}$ gallons of paint?

Step 1

Change mixed numbers to fractions.

$$6\frac{1}{4} \div 2\frac{1}{2} = \frac{25}{4} \div \frac{5}{2}$$

Step 2

Invert the divisor and change the $\div$ to a $\times$.

$$\frac{25}{4} \times \frac{2}{5}$$

Step 3

Cancel if possible and multiply.

$$\frac{\overset{5}{\cancel{25}}}{\underset{2}{\cancel{4}}} \times \frac{\cancel{2}^1}{\cancel{5}_1}$$

Step 4

Change improper fractions to mixed numbers.

$$\frac{5}{2} = 2\frac{1}{2}$$

Bill could paint $2\frac{1}{2}$ sides with $6\frac{1}{4}$ gallons of paint.

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Lesson 5: Operations with Fractions

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